

The self force and gravitational waves from black-hole binary inspirals

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N. Warburton (University of Southampton)

B. Wardell (University of Southampton)

28 July 2024

Introduction

Two problems arise with Newton's law of gravity and Special Relativity:

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- 1 'Action at a distance' is incompatible with finite speeds.

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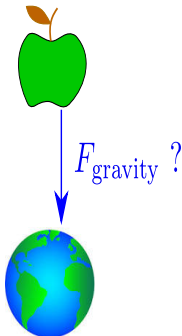
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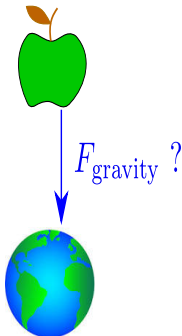
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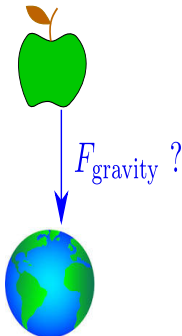


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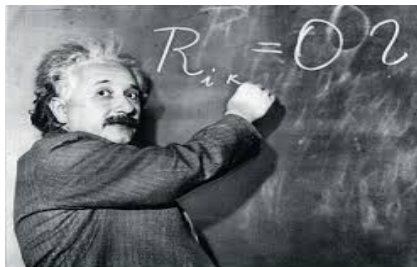
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Einstein searched for a
FIELD THEORY OF GRAVITY
analogous to Faraday & Maxwell's model of electromagnetism.

In 1915 Einstein published the theory of General Relativity

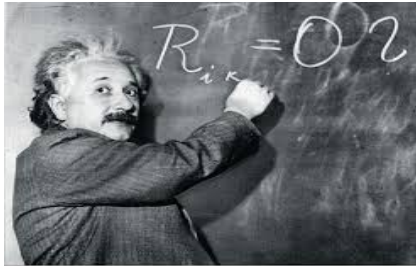
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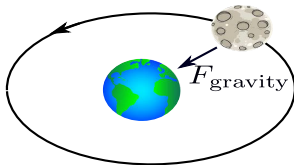
GR is the discovery that spacetime and the gravitational field are the same thing!

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Instead of Newton's picture of gravity as a 'force acting at a distance'

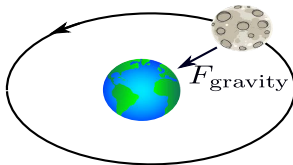
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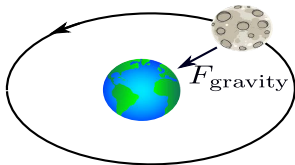
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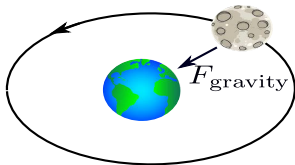


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Massive objects produce a curvature of the spacetime.

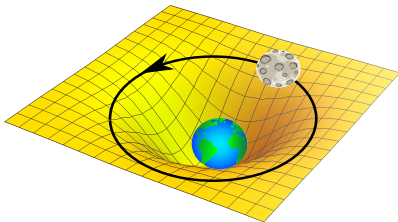
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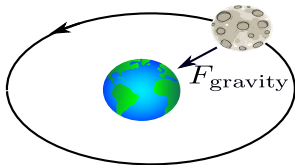
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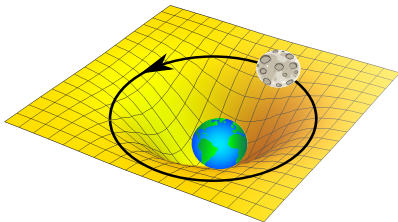
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The gravitational field is the geometric shape of the spacetime.

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General relativity predicts the existence of **Gravitational Waves** ...

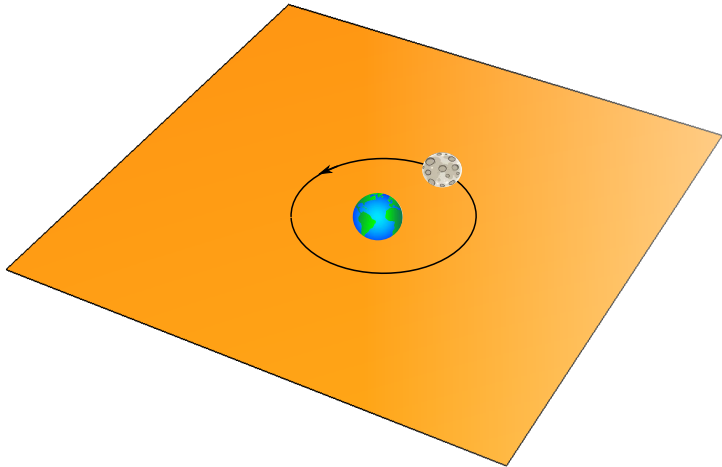
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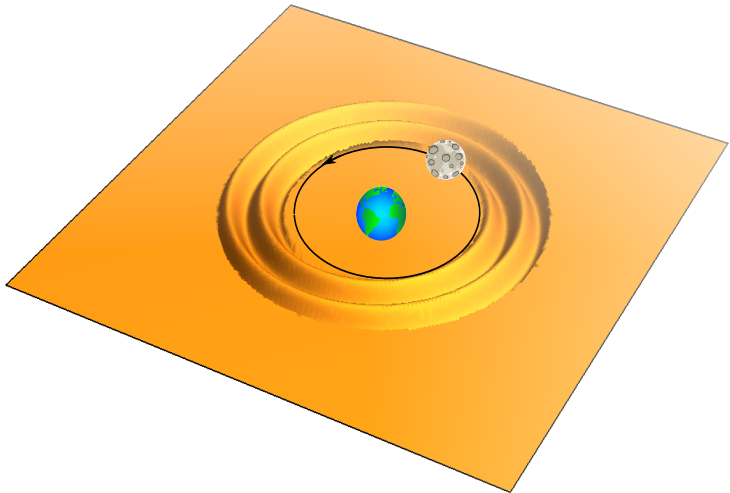
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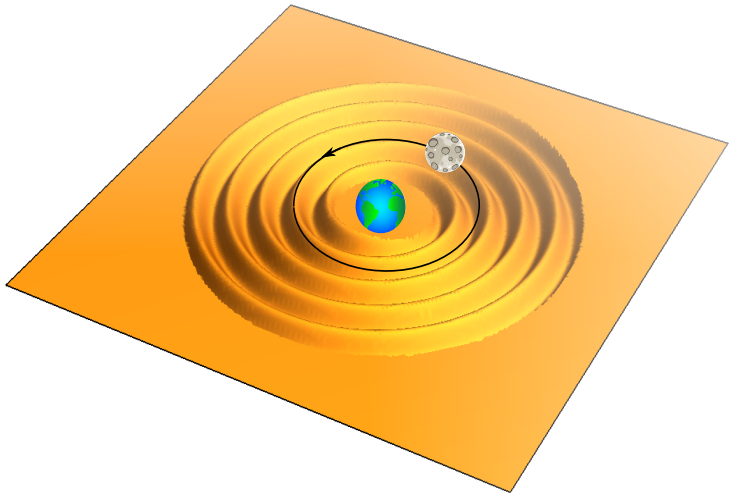
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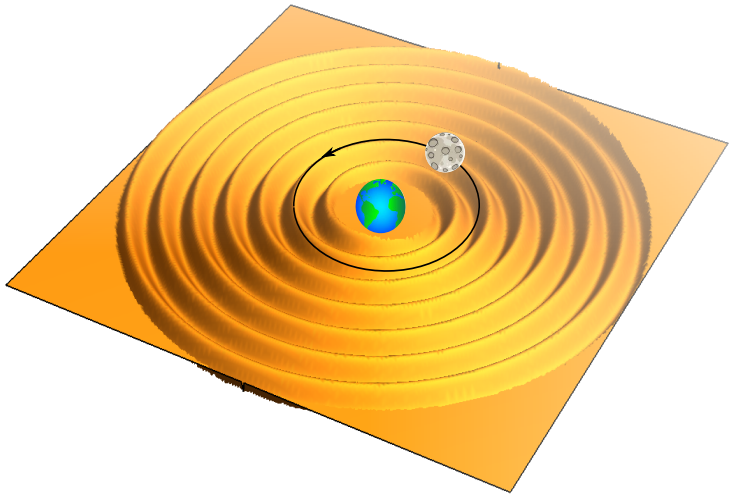
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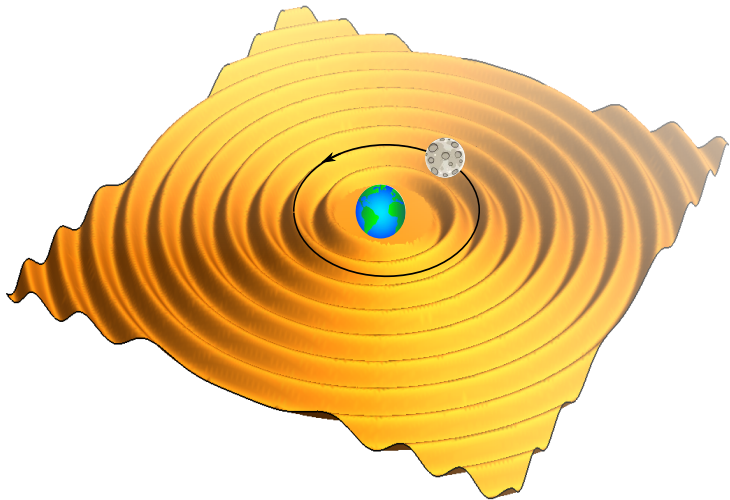
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The first detection of a gravitational wave

In 2015, in a black hole binary far far away...

The first detection of a gravitational wave

**GRAVITATIONAL
WAVES DETECTED!**

The first detection of a gravitational wave

Episode IX

THE DAWN OF GRAVITATIONAL WAVE ASTRONOMY

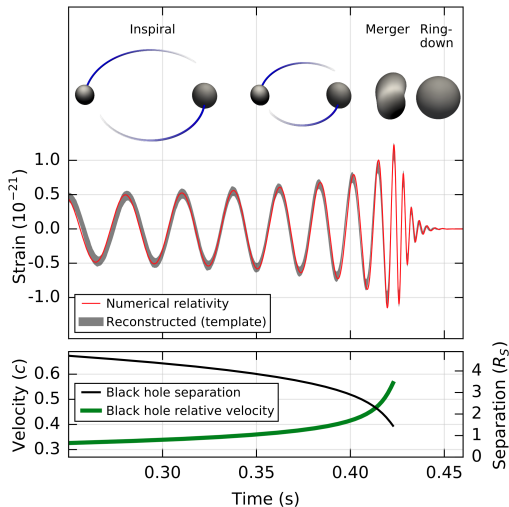
*In 2015, nearly 100 years after
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first gravitational wave from a
black hole binary merger. This
marks the beginning of a new era:
gravitational wave astronomy*

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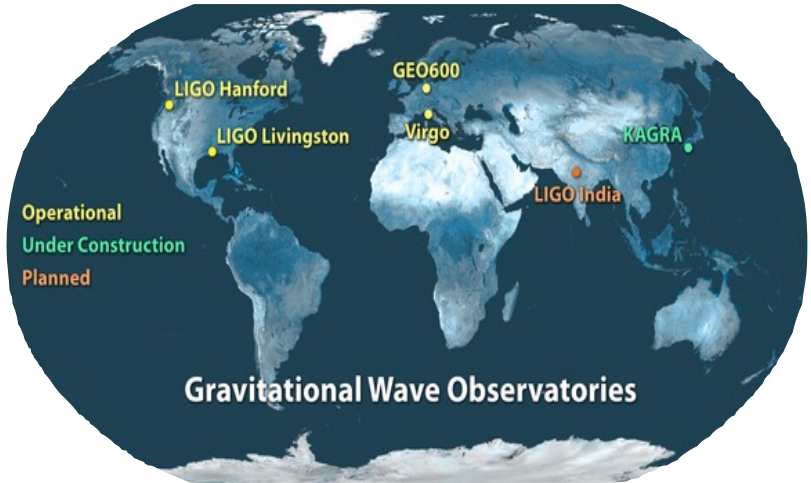
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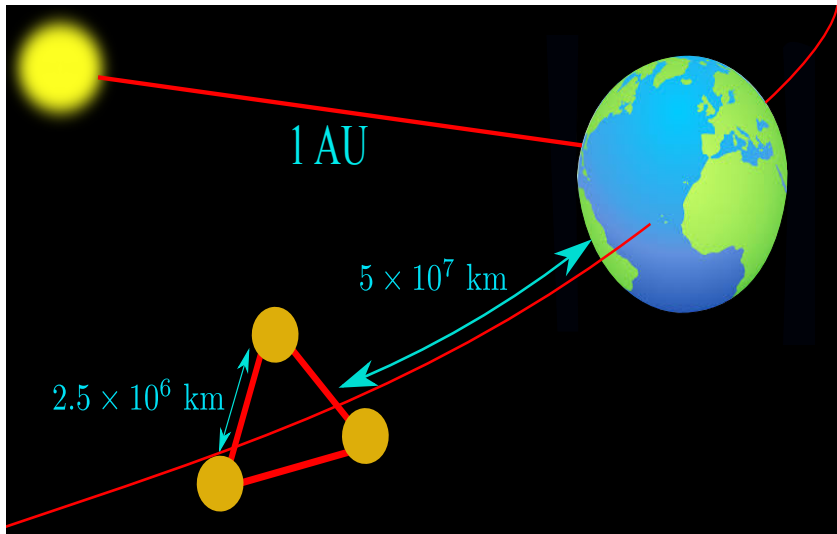
Gravitational wave detectors

- Today a number of GW detectors are in various stages of development on the ground...



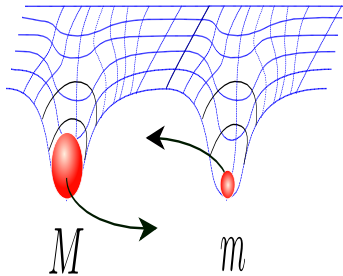
Gravitational wave detectors

- and in space, including DECIGO and LISA.



Binary inspirals

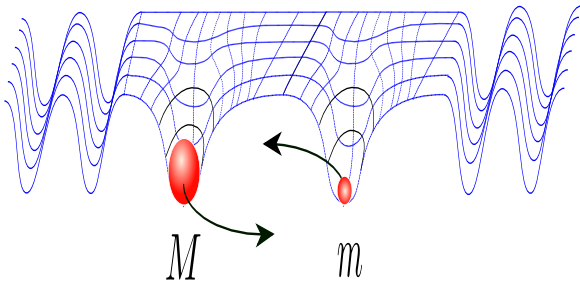
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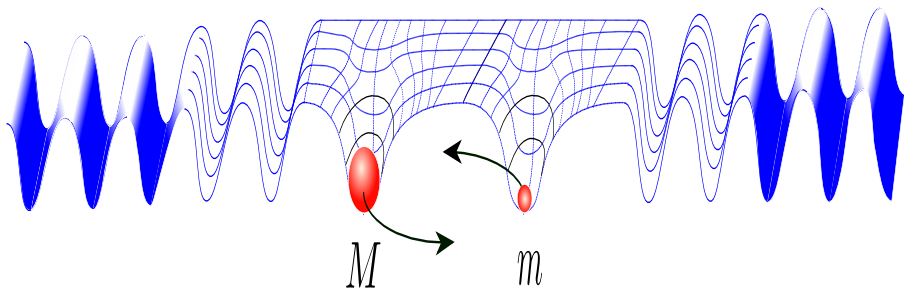


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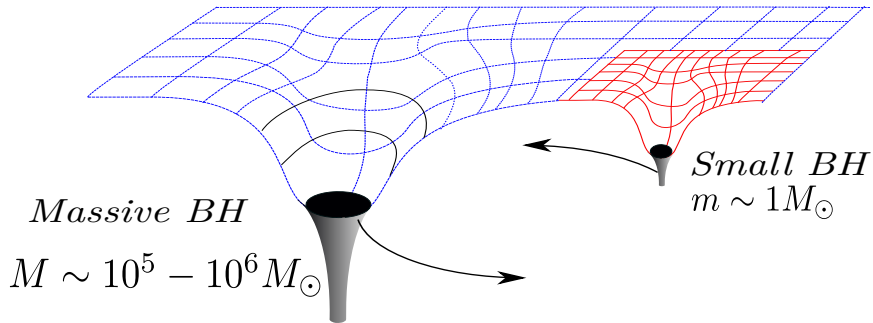
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The waves can be detected on Earth.



Extreme mass ratio inspirals



Small mass ratio,

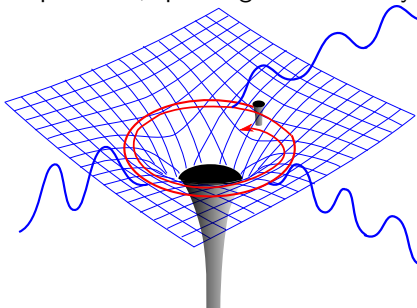
$$\epsilon := \frac{m}{M} \ll 1.$$

EMRI GWs are the most information rich

The small BH moves on a tight orbit ~ 1 yr, tracing a detailed map of the spacetime, spanning $\sim 10^5$ GW cycles!

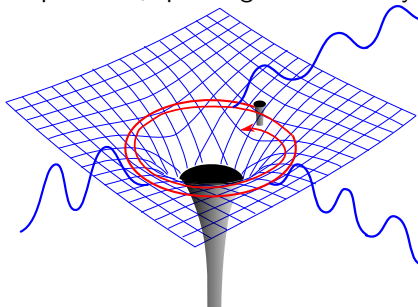
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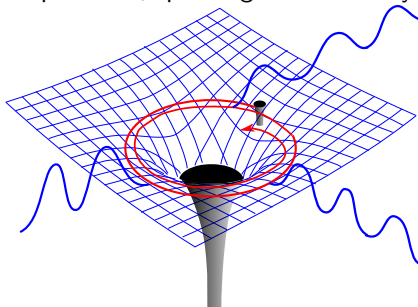
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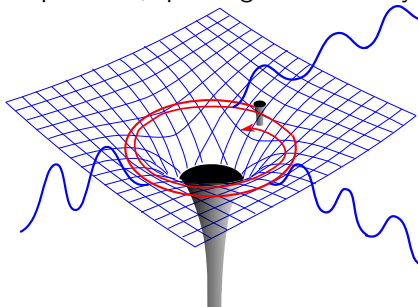


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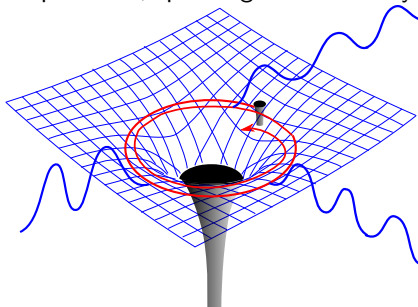


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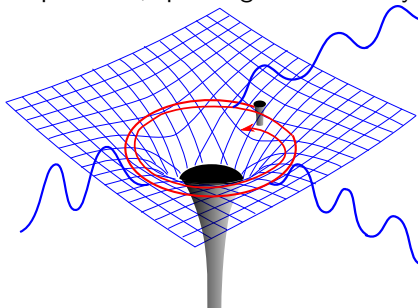


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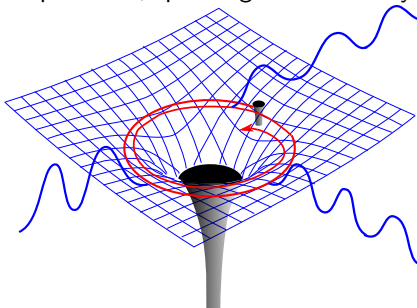


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- And not to mention a direct test of ...

No hair theorem!

... the no hair theorem!



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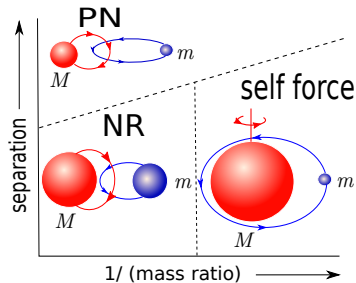
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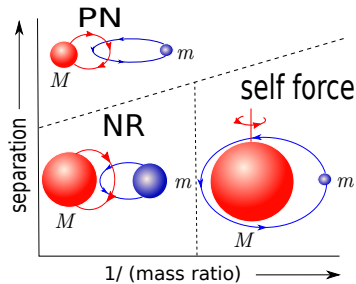
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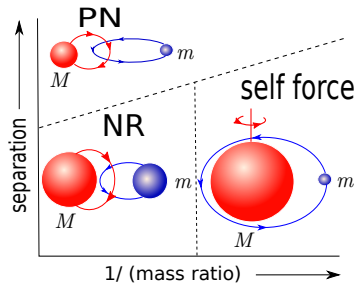


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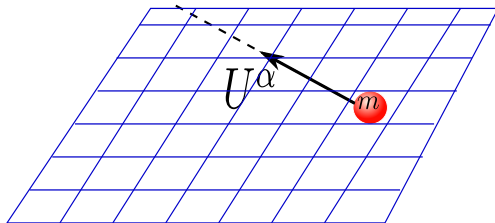
gravitational self-force theory!

Geodesic motion

In flat space test particles move in straight lines

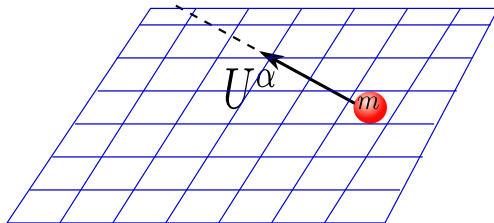
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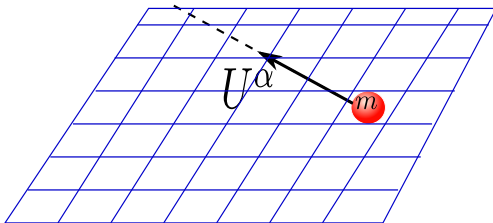
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Four velocity U^α is a tangent of the worldline.

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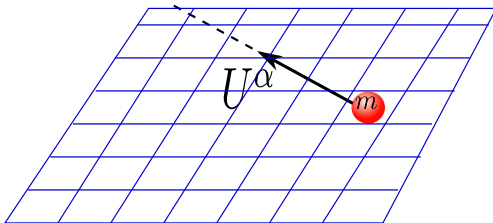


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Equation of motion:

Geodesic motion

In flat space test particles move in straight lines



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Equation of motion:

$$\frac{dU^\alpha}{d\tau} = 0 .$$

Geodesic motion

Einstein equations

Geodesic motion

Einstein equations $\Rightarrow$

Geodesic motion

Einstein equations $\Rightarrow$ Equivalence principle:

Geodesic motion

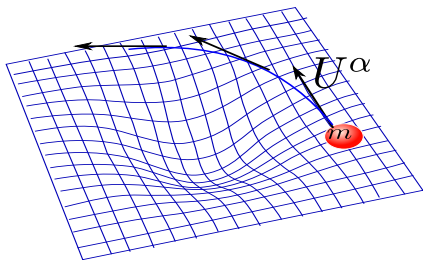
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In curved space test particles move on **geodesics of the background spacetime**

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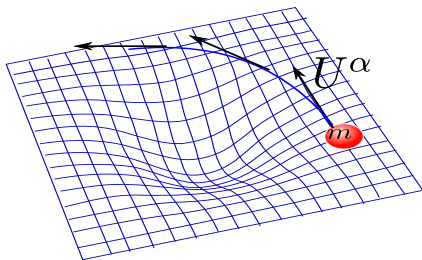
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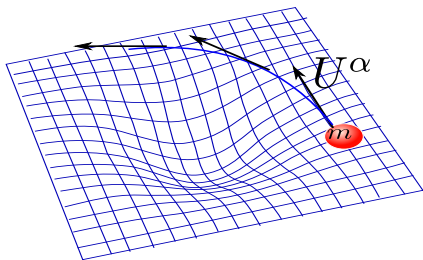


A geodesic is a trajectory where 4 velocity is always tangent to the worldline.

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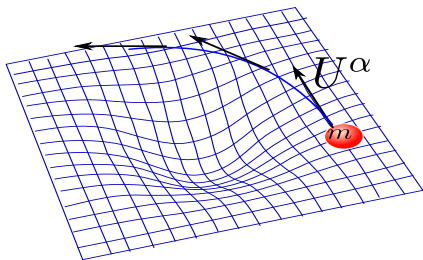
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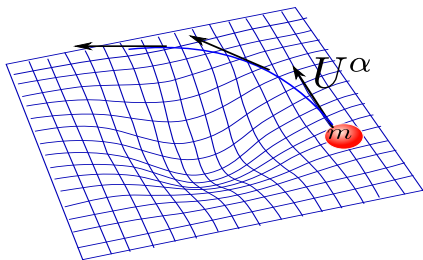
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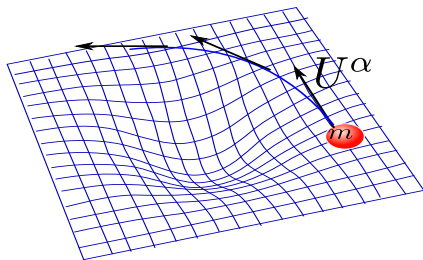
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Covariant derivative:

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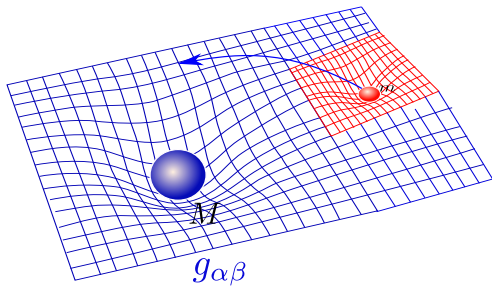
Covariant derivative:
$$\frac{D[g]U^\alpha}{d\tau} := \frac{dU^\alpha}{d\tau} + \Gamma(g)^\alpha_{\mu\nu} U^\mu U^\nu .$$

The gravitational self-force

Take a binary system of masses M and m .

The gravitational self-force

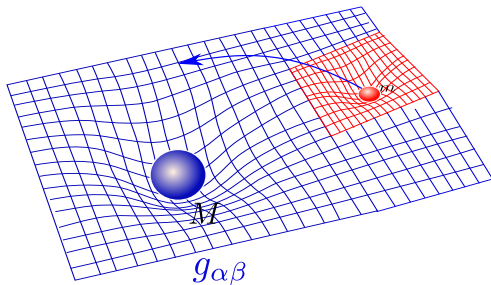
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Let g be the metric induced by M , ignoring the field of m .

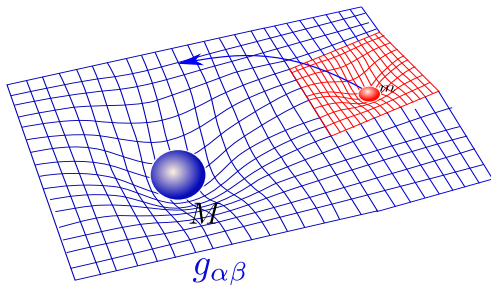


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Then m would move on a geodesic of the background spacetime g .



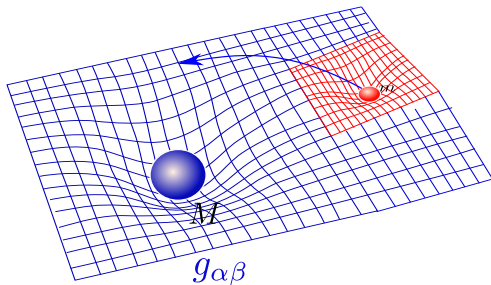
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Now include the effect of the field h of the small mass itself,



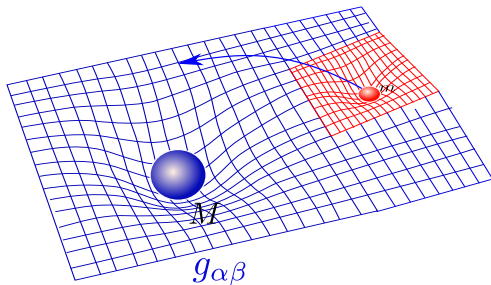
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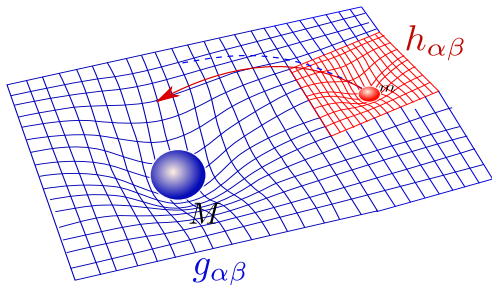
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Then m would move on a geodesic of the background spacetime g .

Now include the effect of the field h of the small mass itself, m deviates from geodesic motion in the background!



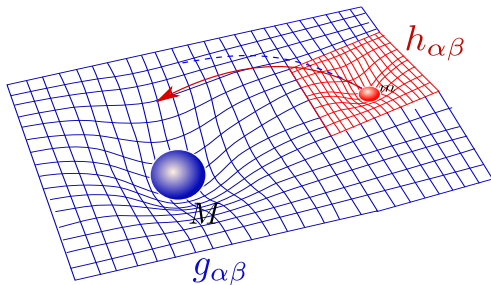
The gravitational self-force

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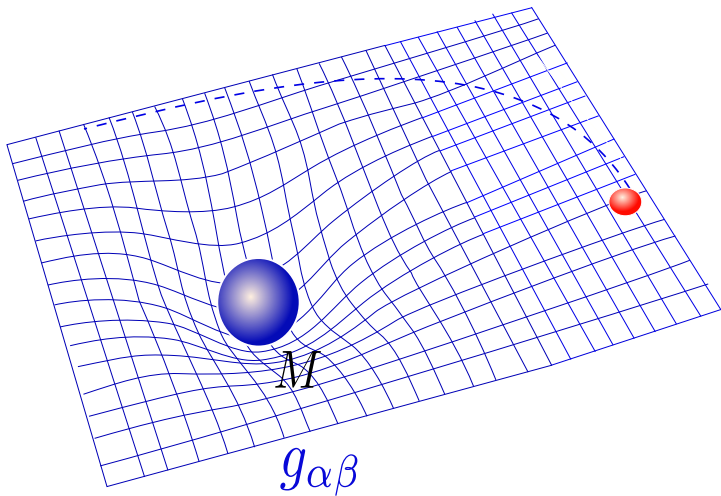
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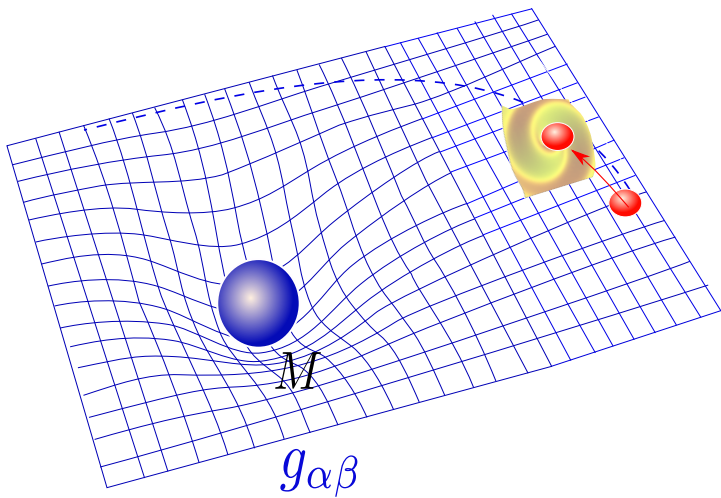
This very acceleration from geodesic motion is precisely the effect of the self force!

The gravitational self-force



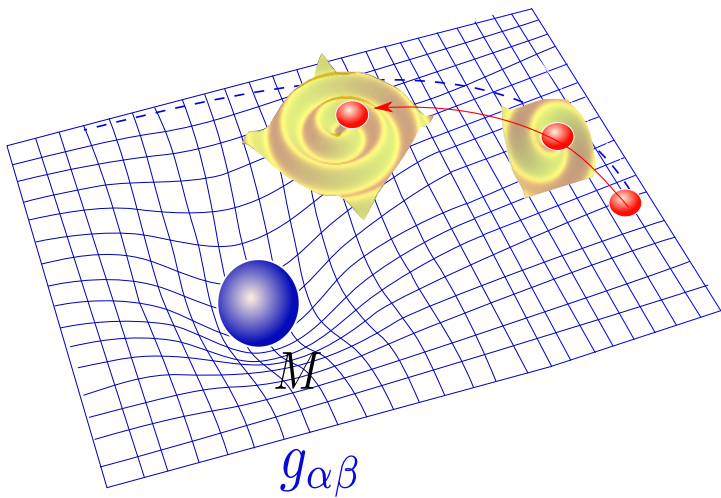
The gravitational self-force

The self force accelerates a test particle away from geodesic motion.



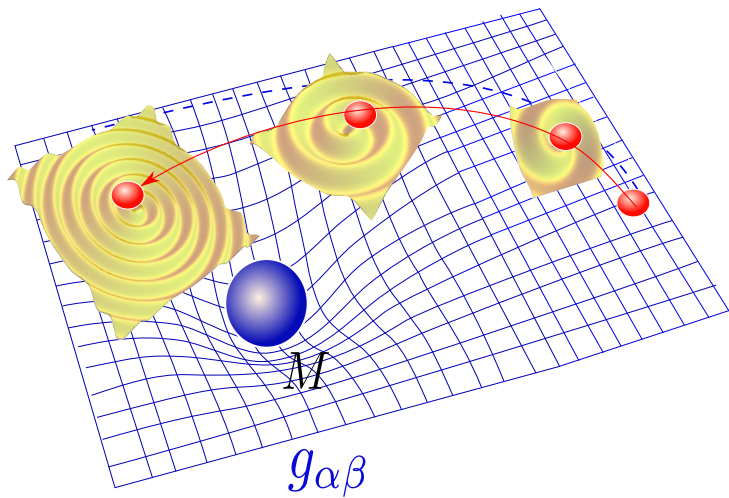
The gravitational self-force

Ergo a test particle radiates energy through gravitational waves.

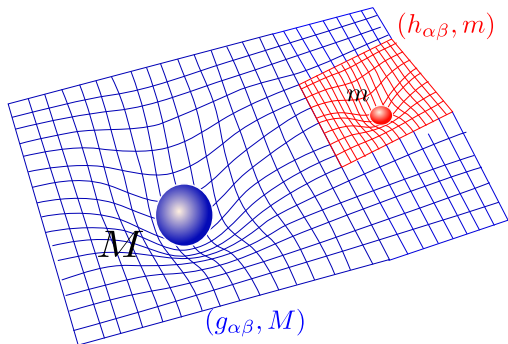


The gravitational self-force

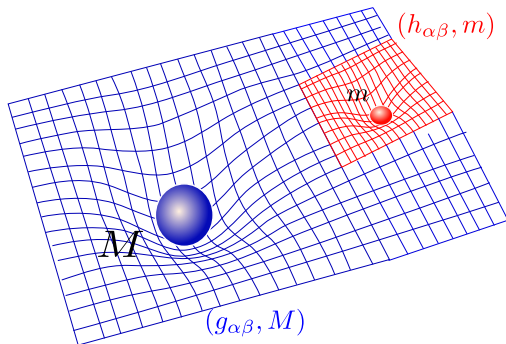
The dissipative self-force is responsible for gravitational radiation.



Matched asymptotic expansions

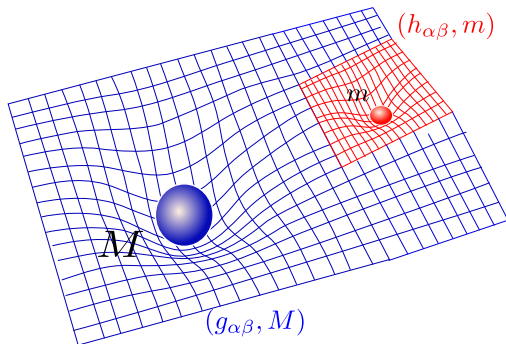


Matched asymptotic expansions



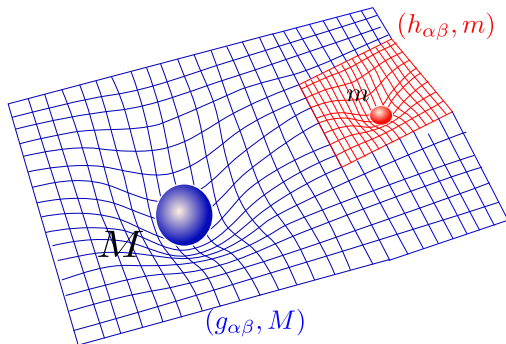
For small mass ratios,

Matched asymptotic expansions



For small mass ratios, $\epsilon \equiv \frac{m}{M} \ll 1$.

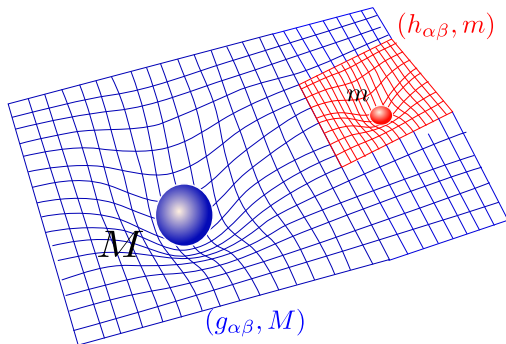
Matched asymptotic expansions



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The field $h_{\alpha\beta}$ of m is treated as a small perturbation to the field $g_{\alpha\beta}$.

Matched asymptotic expansions

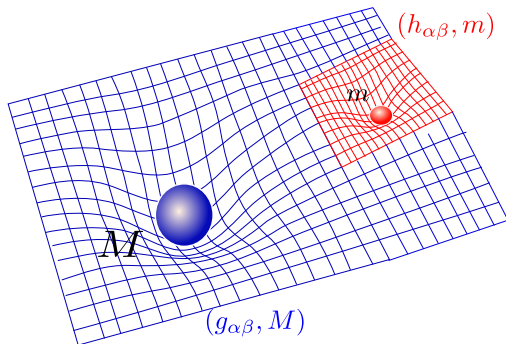


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Matched asymptotic expansions



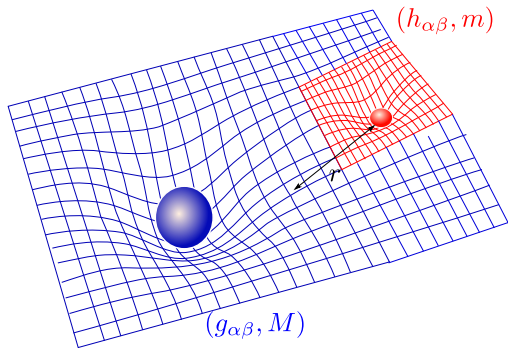
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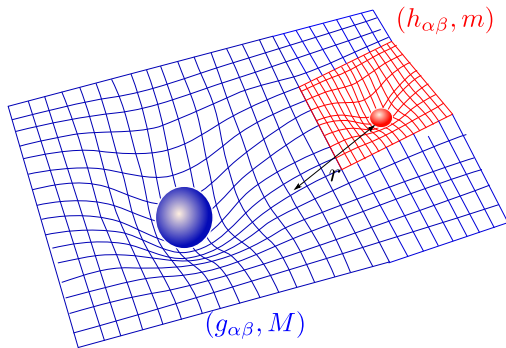
The metric of the full spacetime is expanded in powers of ϵ :

$$\tilde{g}_{\alpha\beta} = g_{\alpha\beta} + \epsilon h_{\alpha\beta}^{(1)} + \epsilon^2 h_{\alpha\beta}^{(2)} + O(\epsilon^3).$$

Matched asymptotic expansions

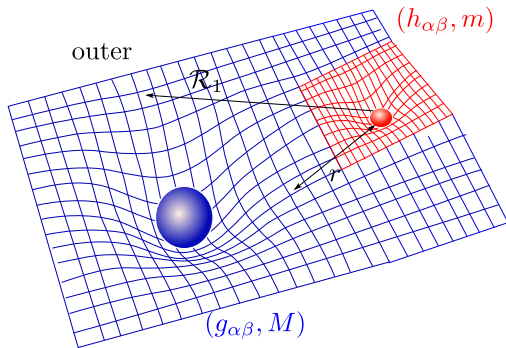


Matched asymptotic expansions



Expand $h_{\alpha\beta}^{(1)}$ and $h_{\alpha\beta}^{(2)}$ in powers of r in 2 different regions:

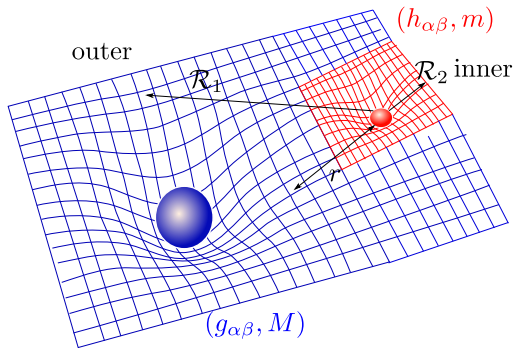
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Expand $h_{\alpha\beta}^{(1)}$ and $h_{\alpha\beta}^{(2)}$ in powers of r in 2 different regions:

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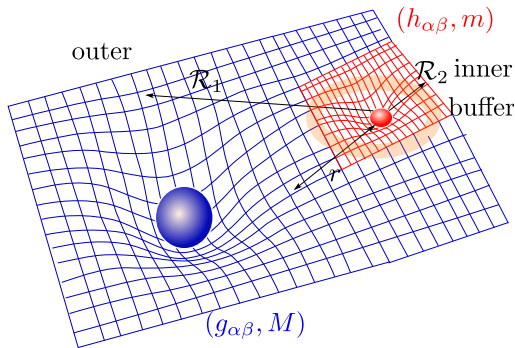
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Matched asymptotic expansions

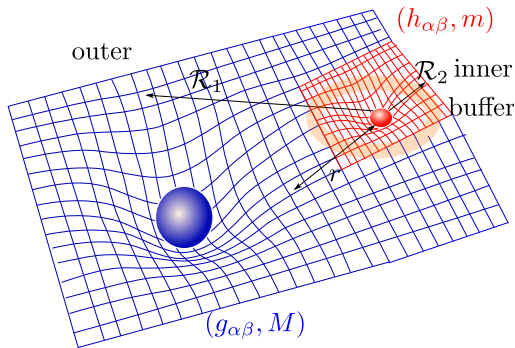


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Match the two expansions order-by-order in ϵ and r in the buffer region.

Singular - regular field

$$h_{\alpha\beta}^{(1)}$$

Singular - regular field

$$h_{\alpha\beta}^{(1)} = \frac{1}{r} h_{\alpha\beta}^{1,-1}$$

Singular - regular field

$$h_{\alpha\beta}^{(1)} = \frac{1}{r} h_{\alpha\beta}^{1,-1} + h_{\alpha\beta}^{1,0} + r h_{\alpha\beta}^{1,1} + r^2 h_{\alpha\beta}^{1,2} + \dots$$

Singular - regular field

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Singular - regular field

singular

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Singular - regular field

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Singular - regular field

singular

regular

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Singular - regular field

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Singular - regular field

singular

regular

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Singular - regular field

singular

regular

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$$h_{\alpha\beta}^{(2)} = h_{\alpha\beta}^{(S2)} + h_{\alpha\beta}^{(R2)}$$

Singular - regular field

singular

regular

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Singular - regular field

singular

regular

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Singular - regular field

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Self force depends entirely on the regular field.

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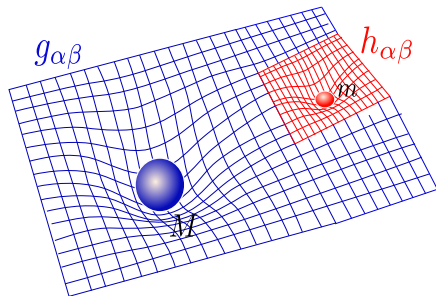
Singular field does not affect the motion.

The generalized equivalence principle

Consider a binary system.

The generalized equivalence principle

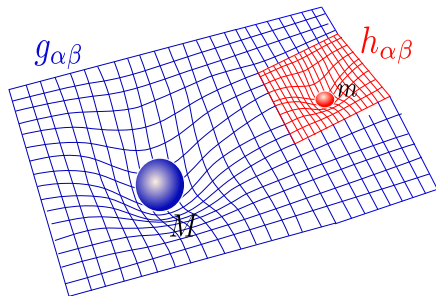
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The generalized equivalence principle

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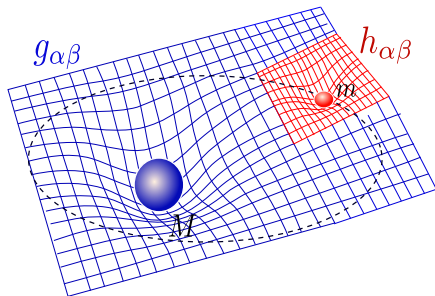


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- At leading order, the small object moves on a geodesic of $g_{\alpha\beta}$.

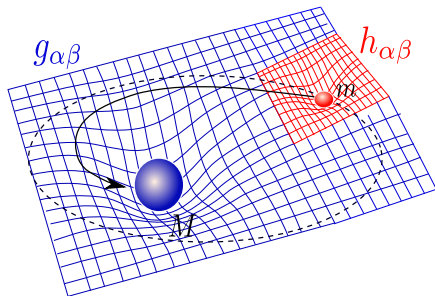


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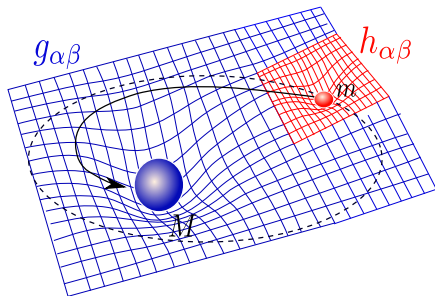
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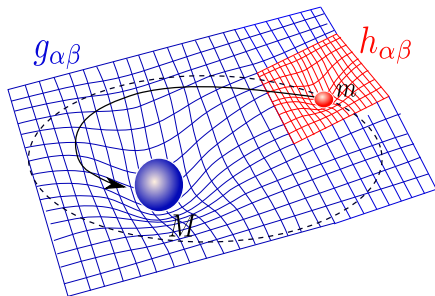
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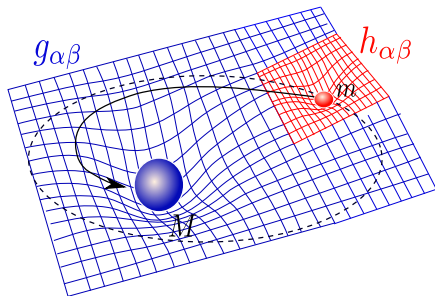
In the *effective spacetime* $g_{\alpha\beta} + h_{\alpha\beta}^R$:

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$\Rightarrow$ acceleration in $g_{\alpha\beta}$ is equivalent to geodesic motion in $g_{\alpha\beta} + h_{\alpha\beta}$.

Brief history of the self force

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$$\Delta t_{\text{inspiral}} = \frac{E}{\dot{E}} \sim \frac{1}{\epsilon}$$

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Ignoring 2nd-order corrections over the inspiral

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$\therefore$ to accurately model EMRIs 2nd-order results are necessary and sufficient.

Black hole perturbation theory

Einstein equations for the full spacetime:

Black hole perturbation theory

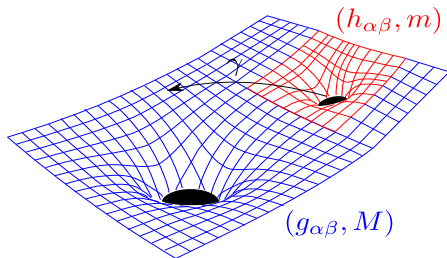
Einstein equations for the full spacetime:

$$G_{\alpha\beta}[\tilde{g}] = 8\pi\tilde{T}_{\alpha\beta}, \quad \tilde{T}_{\alpha\beta}^{\text{point}} = \int_{\gamma} d\tilde{\tau} \tilde{u}_{\alpha} \tilde{u}_{\beta} \delta^4(x-z) (-\tilde{g})^{-1/2}$$

Black hole perturbation theory

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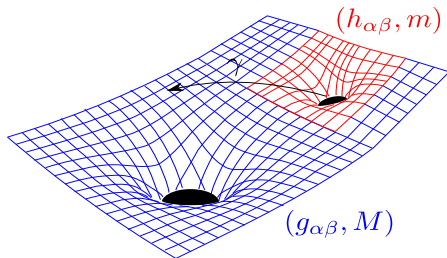
$$G_{\alpha\beta}[\tilde{g}] = 8\pi\tilde{T}_{\alpha\beta}, \quad \tilde{T}_{\alpha\beta}^{\text{point}} = \int_{\gamma} d\tilde{\tau} \tilde{u}_{\alpha} \tilde{u}_{\beta} \delta^4(x-z) (-\tilde{g})^{-1/2}$$



Black hole perturbation theory

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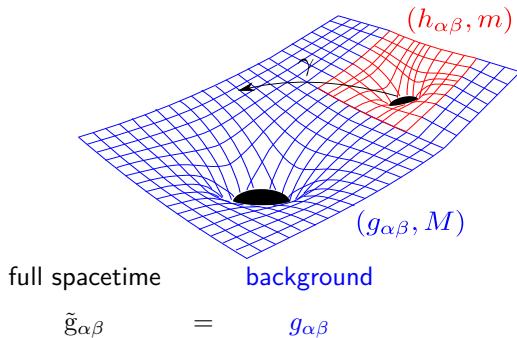
full spacetime

$$\tilde{g}_{\alpha\beta} =$$

Black hole perturbation theory

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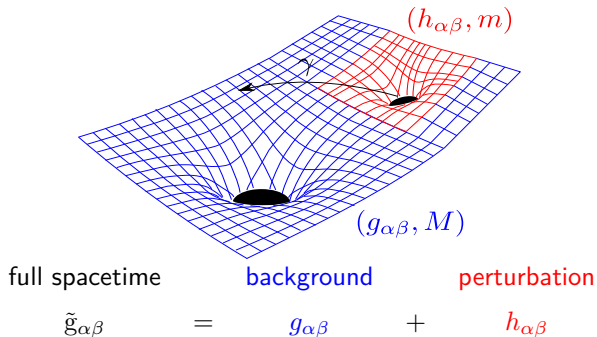
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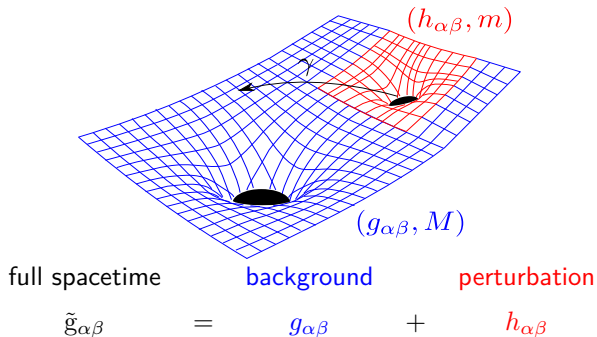
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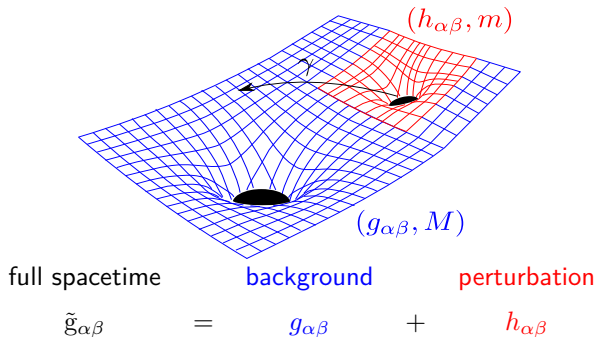


Separate the perturbation into contributions at each order of ϵ :

Black hole perturbation theory

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Separate the perturbation into contributions at each order of ϵ :

$$h_{\alpha\beta} = \epsilon^1 h_{\alpha\beta}^1 + \epsilon^2 h_{\alpha\beta}^2 + O(\epsilon^3), \quad \tilde{T}_{\alpha\beta} = \epsilon^1 T_{\alpha\beta}^1 + \epsilon^2 T_{\alpha\beta}^2 + O(\epsilon^3).$$

$$\mathcal{O}(\epsilon^0)$$

Perturbed Einstein equations

$$O(\epsilon^0) \quad G_{\alpha\beta}[g] = 0$$

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- $\therefore$ we need a new approach for 2nd order...

In 2014, Miller,

Pound, Warburton and Wardell
solved this long standing problem...



...using a 'Puncture scheme'!

The puncture scheme

- The Vacuum field equations

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$$\delta G_{\alpha\beta} [h^1] = 0$$

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- which can be analytically continued down to the worldline!

Circular orbits in Schwarzschild - 1st order

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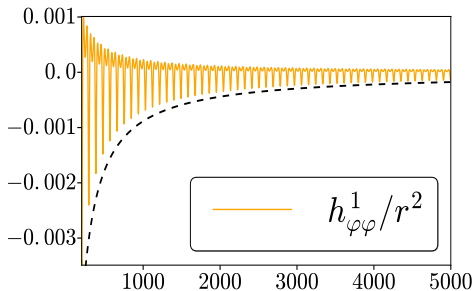
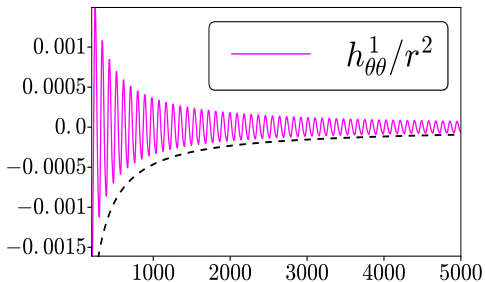
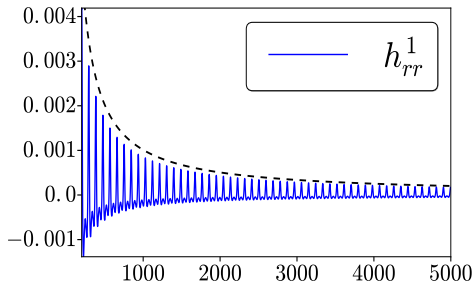
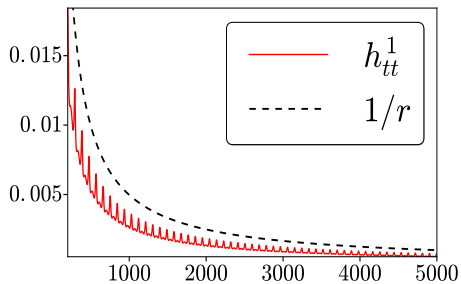
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- We take a “snapshot” $t = t_0$

1st order results



1st order self force

r_0/M	F_1^t (Miller 2016)	F_1^t Akcay 2012	rel. diff.
6.0	$-1.99470434(7) \times 10^{-3}$	$-1.9947610064(3) \times 10^{-3}$	10^{-6}
10.0	$-9.19105764(4) \times 10^{-5}$	$-9.19075772(7) \times 10^{-5}$	10^{-5}
20.0	$-2.2554354(8) \times 10^{-6}$	$-2.2554391(2) \times 10^{-6}$	10^{-6}

r_0/M	F_1^r (Miller 2016)	F_1^r Akcay 2012	rel. diff.
6.0	$2.4466467(3) \times 10^{-3}$	$2.4466495(4) \times 10^{-3}$	10^{-6}
10.0	$1.3389384(1) \times 10^{-2}$	$1.3389470(2) \times 10^{-2}$	10^{-5}
20.0	$4.1570134(6) \times 10^{-3}$	$4.1570550(2) \times 10^{-3}$	10^{-5}

Moving to 2nd order

$$\delta G_{\alpha\beta} [h^{R1}] = -\delta G_{\alpha\beta} [h^{S1}]$$



$$\delta G_{\alpha\beta} [h^{R1}] = -\delta G_{\alpha\beta} [h^{S1}] \quad \checkmark$$

$$\delta G_{\alpha\beta} [h^{R2}] = -2\delta^2 G_{\alpha\beta} [h^1, h^1] - \delta G_{\alpha\beta} [h^{S2}] \quad ?$$

How to calculate the 2nd order source (PRD 2015)

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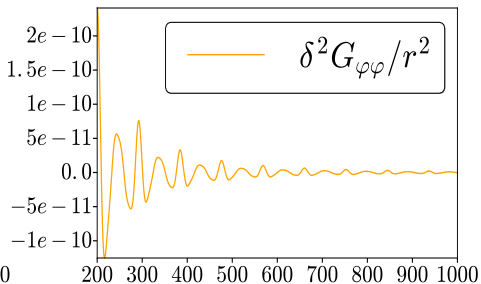
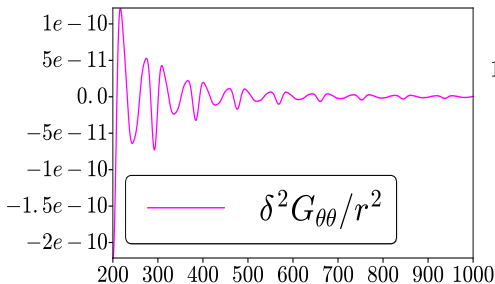
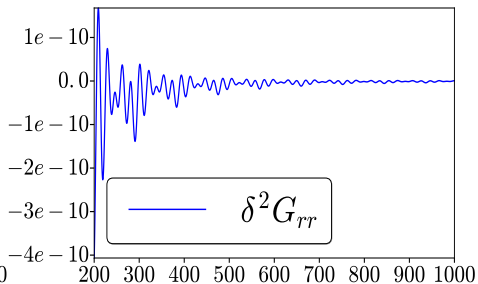
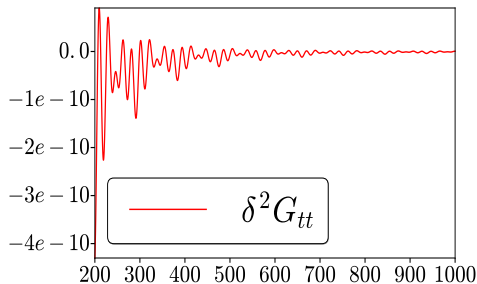
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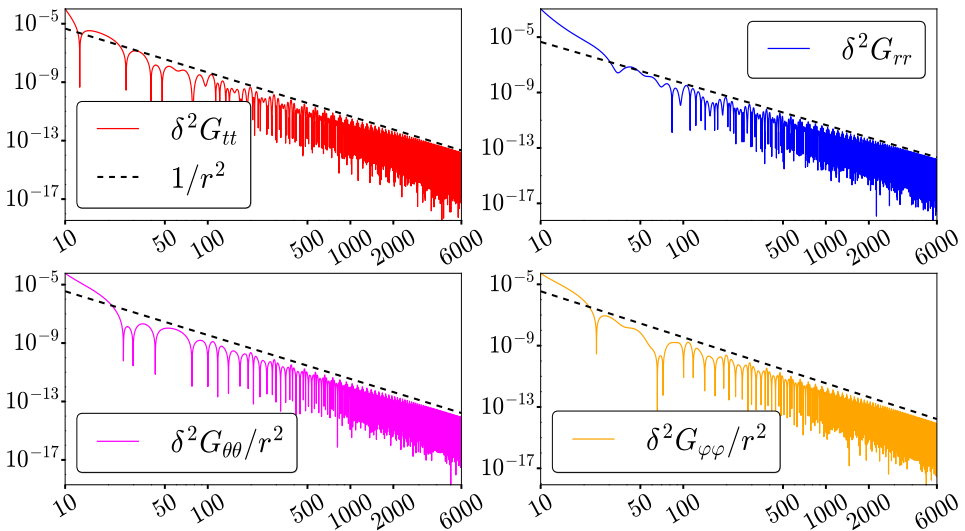
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2nd order source (PRD 2015)



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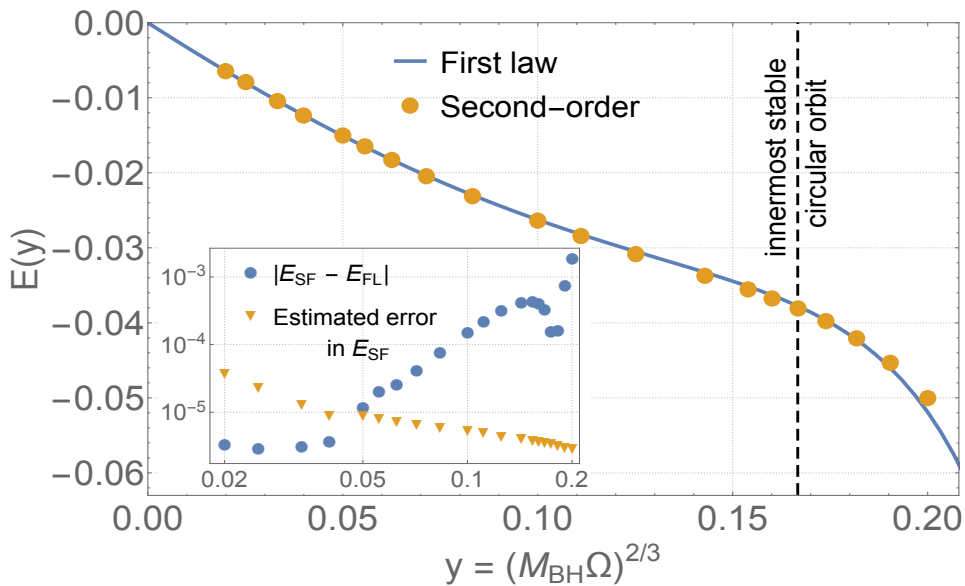


Binding energy - first calculation of a 2nd order effect (PRL 2018)

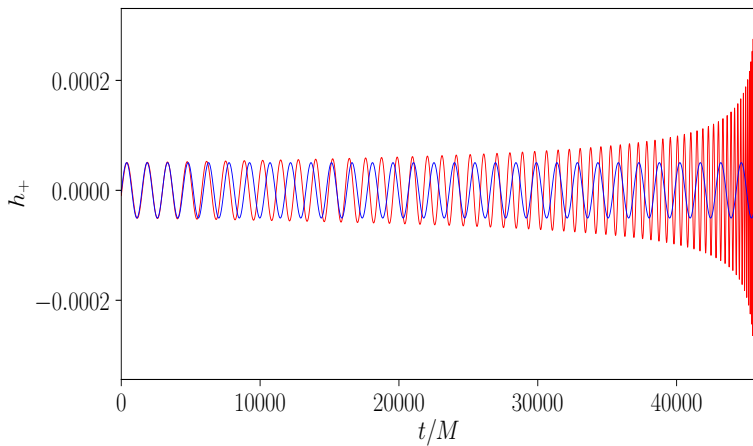
$$E_{\text{Binding}} := \frac{M_{\text{Bondi}} - M_{\text{BH}} - m}{\nu}$$

$$\nu := \frac{m M_{\text{BH}}}{M_{\text{BH}} + m}$$

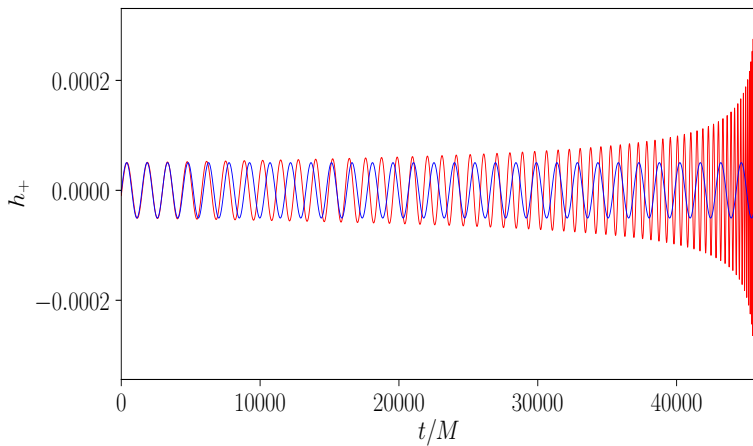
Binding energy - comparison with the first law (PRL 2019)



Waveform evolution with time (PRD 2021)

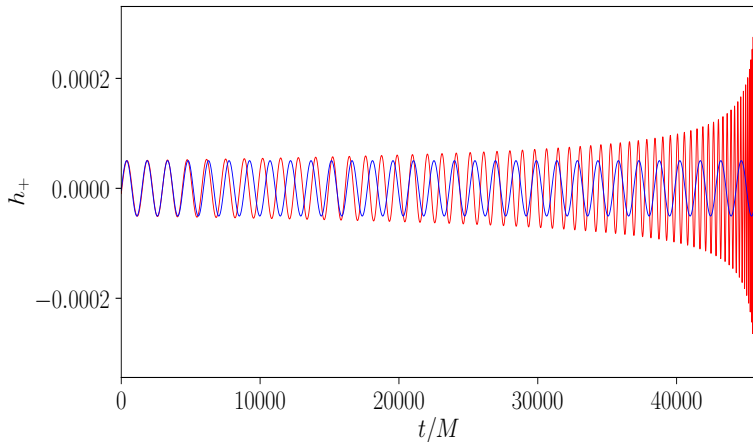


Waveform evolution with time (PRD 2021)



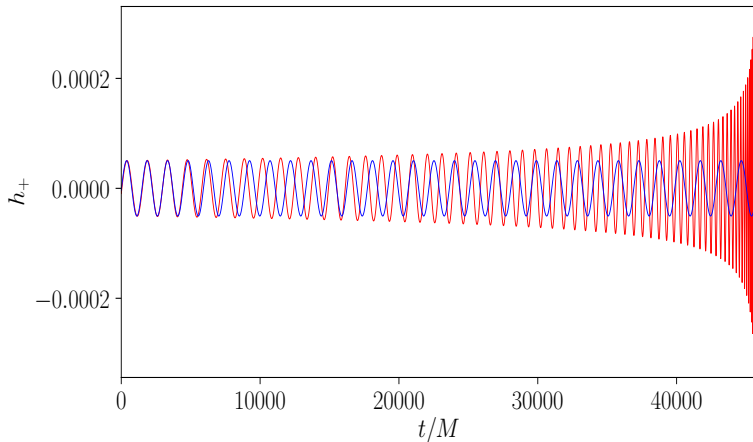
Thin red curve:

Waveform evolution with time (PRD 2021)



Thin red curve: waveform from an adiabatic orbit
for a mass ratio $\epsilon = 0.1$.

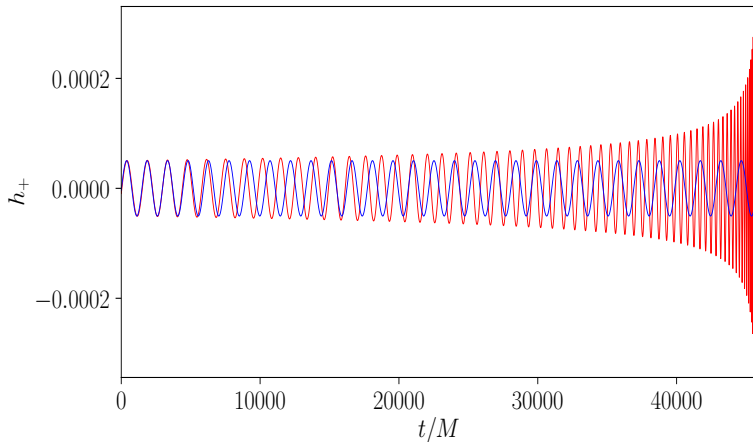
Waveform evolution with time (PRD 2021)



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Blue curve:

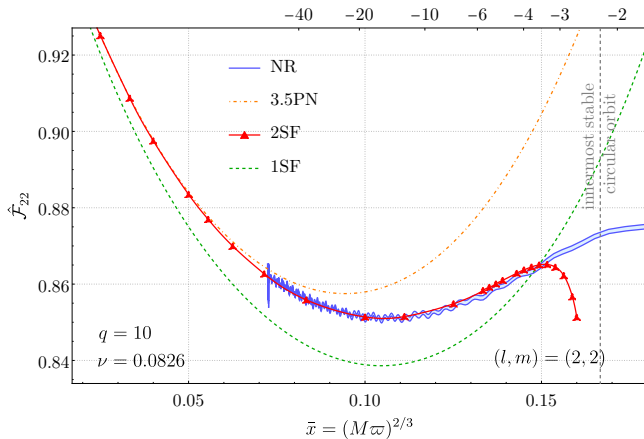
Waveform evolution with time (PRD 2021)



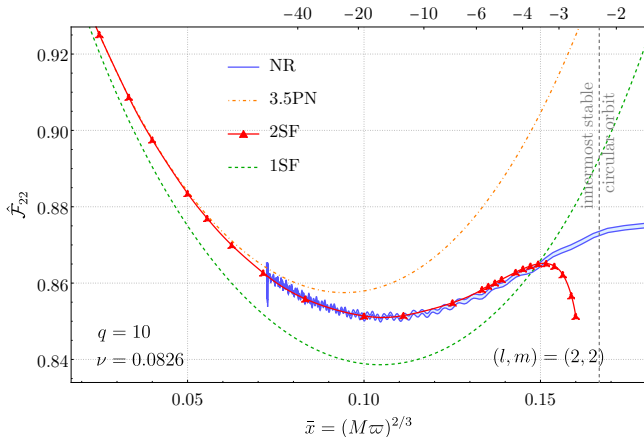
Thin red curve: waveform from an adiabatic orbit
for a mass ratio $\epsilon = 0.1$.

Blue curve: waveform from a point mass on a circular geodesic
with the same initial frequency.

Gravitational-wave energy flux(PRL 2021)

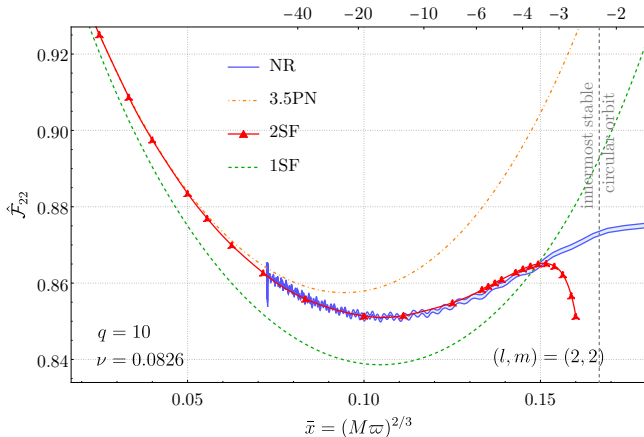


Gravitational-wave energy flux(PRL 2021)



GW flux for a binary as a function of inverse orbital separation using PN, NR and GSF approaches. Numbers on the top axis count the cycles before the peak amplitude in the NR waveform.

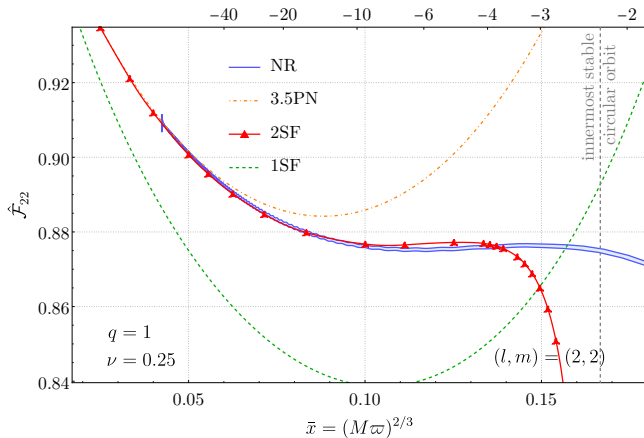
Gravitational-wave energy flux(PRL 2021)



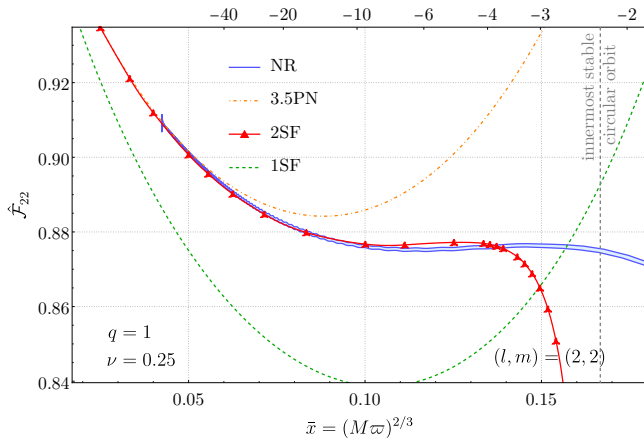
GW flux for a binary as a function of inverse orbital separation using PN, NR and GSF approaches. Numbers on the top axis count the cycles before the peak amplitude in the NR waveform.

Our 2nd order GSF calculation agrees well with the NR result until very close to merger, where GSF contributions diverge as the two-timescale approximation breaks down.

Gravitational-wave energy flux(PRL 2021)

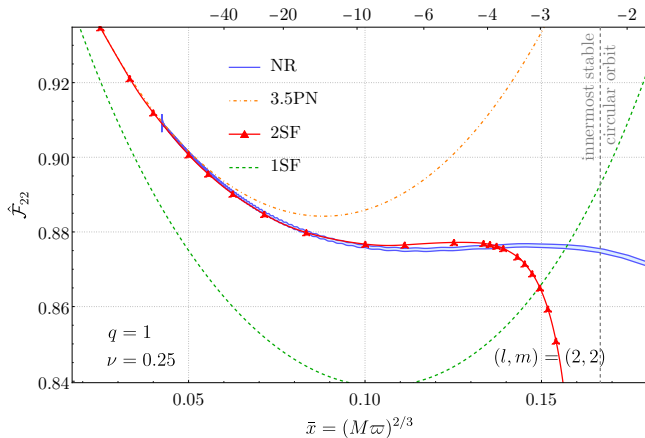


Gravitational-wave energy flux(PRL 2021)



Same as above but with $q = 1$

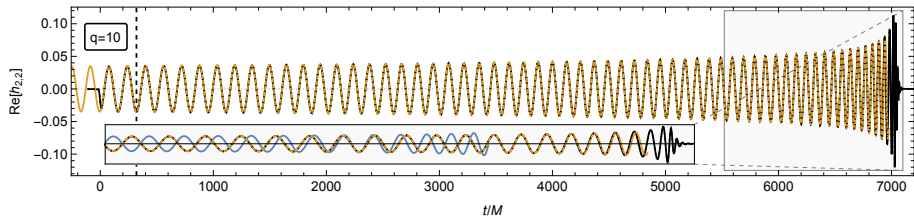
Gravitational-wave energy flux(PRL 2021)



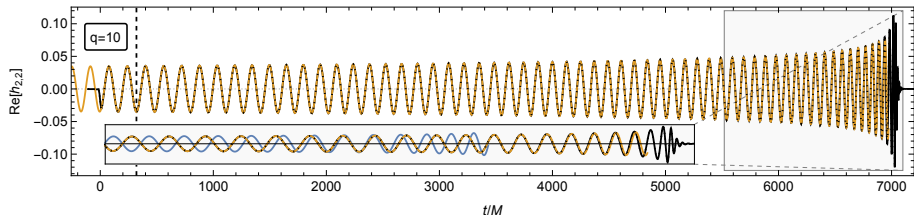
Same as above but with $q = 1$

Despite being a small- ϵ (small- ν , large- q) expansion, the 2SF result agrees remarkably well with the NR flux for this equal-mass binary.

Gravitational wave form(PRL 2021)

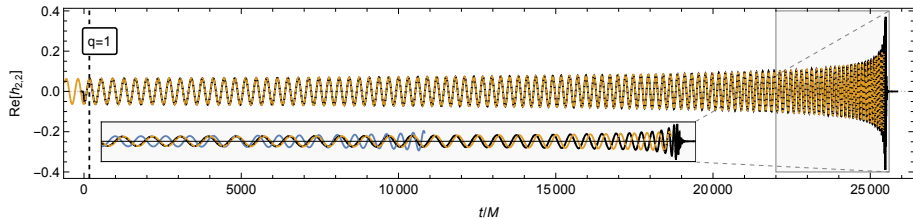


Gravitational wave form(PRL 2021)

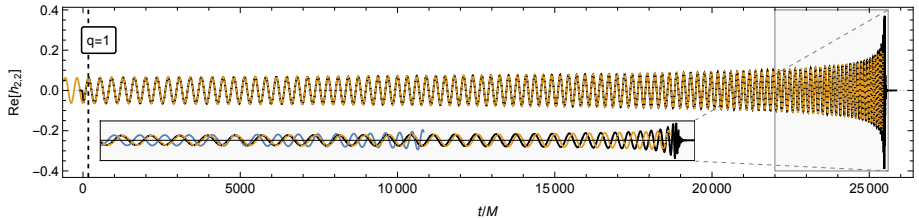


1st post-adiabatic order (1PA) time-domain Waveform for a mass ratio 1 : 10 nonspinning binary (orange) and waveform for the same binary using NR (black). The waveforms are aligned in time and phase at $t = 320M$, $1/x \approx 13.83$.

Gravitational wave form(PRL 2021)

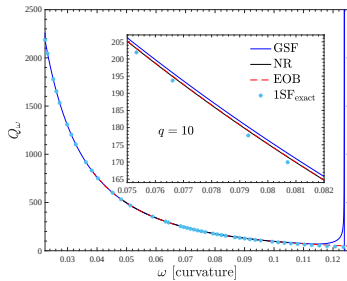
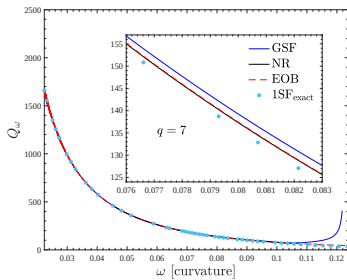


Gravitational wave form(PRL 2021)

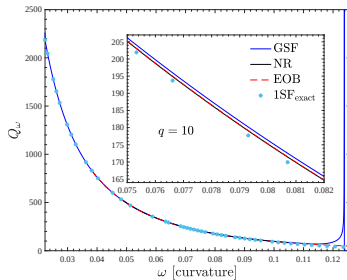
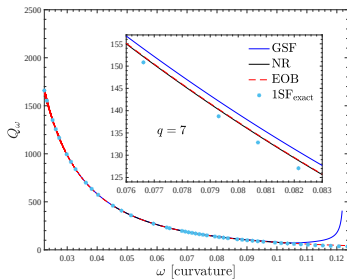


1PA time-domain waveform for a mass ratio 1 : 1 nonspinning binary (orange) and NR calculation (black) for the same binary. The waveforms are aligned in time and phase at $t = 170M$, where $1/x \approx 23.94$.

Q function (PRL 2022)

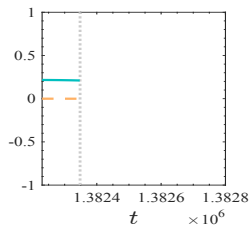
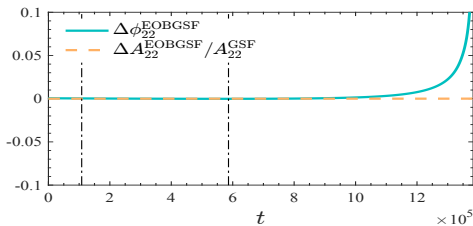
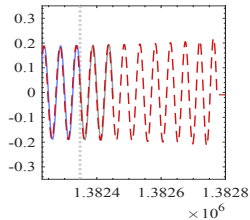
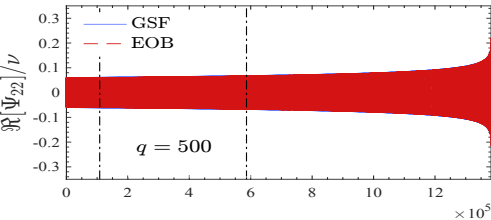


Q function (PRL 2022)

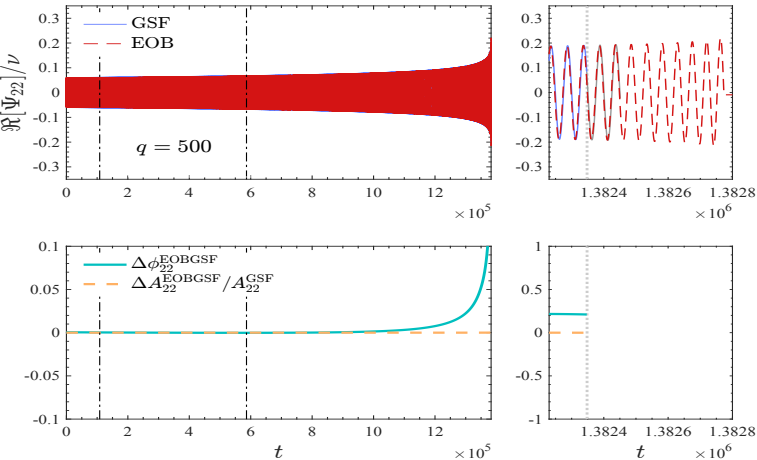


NR, GSF and EOB comparison of Q_ω functions obtained from the phase of ψ_4^{22} .

Q function (PRL 2022)



Q function (PRL 2022)



EOB/GSF phasing for a $q = 500$ binary. Vertical dash-dotted lines indicate times corresponding to the $[\omega_L, \omega_R]$ alignment interval. Dotted line corresponds to $\omega_{22}^{\text{GSF break}}$.

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- Computation of the binary Q function up to 2nd order and successful comparison with PN, NR and EOB results.

Questions...

